

Radiologically Relevant Clinical History Summarization with Large Language Models: A Multireader Performance Study

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See also the editorial by Yilmaz and Cardoza-Ochoa in this issue.

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Background: Clinical histories accompanying imaging orders guide protocol selection and diagnostic focus. However, they are often incomplete, potentially compromising diagnostic accuracy and workflow efficiency.

Purpose: To evaluate whether large language models (LLMs) can improve the clinical utility of provided imaging indications by leveraging clinical notes.

Materials and Methods: This retrospective study curated a dataset from deidentified electronic health records at the University of California San Francisco (January 2012 to August 2024), consisting of radiology reports with paired referring clinician–provided and radiologist–curated indications linked to clinical notes. The dataset was stratified across five body systems and five pathophysiologic categories to derive LLM selection and reader study internal test sets. For the reader study, 20 radiologists with 2–25 years of experience compared indications from the referring clinician, radiologist, and best-performing LLMs. Readers scored comprehensiveness, factuality, and conciseness and ranked indications for usefulness in protocoling, usefulness in interpretation, and overall ranking. Models and clinicians were compared using cumulative link mixed models with Tukey-adjusted post hoc comparisons.

Results: From 28 313 patients (mean age, 59 years $\pm$ 20.6 [SD]; 14 912 women), 250 examinations from 247 patients were sampled for the reader study. After nine exclusions, 241 examinations were analyzed, yielding 482 reader-examination evaluations. Indications from the best-performing proprietary (Claude 3.5 Sonnet; Anthropic) and open-source (Qwen 2.5-7B Instruct; Alibaba) LLM were rated as more comprehensive (Likert rating of 5: 37.14% and 28.42%, respectively; both $P < .001$) and factual (68.05% and 59.75%; both $P < .001$) than referring clinician indications. The proprietary LLM ranked most useful in protocoling (rank 1: 40.87%; all $P < .001$), useful in interpretation (44.61%; all $P < .001$), and overall ranking (44.19%, all $P < .001$). Comprehensiveness (65.77% of ratings; both $P < .001$) most strongly influenced overall rankings.

Conclusion: LLMs generated radiology-relevant indications from clinical notes that were more comprehensive and factual than clinician indications, and when generated by the proprietary LLM, were ranked most useful in protocoling and imaging interpretation.

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Radiologists rely on clinical information in imaging indications, which communicate key aspects of patients' medical history and the question the imaging examination intends to assess. Indications affect an imaging examination well before the patient is scanned by guiding the selection of the appropriate imaging protocol. Yet despite their importance, provided imaging indications are often inadequate (1–3), partially due to the lack of awareness of their importance by ordering clinicians and increasingly busy clinician workflows. For example, the commonly received indication of “cancer eval” lacks essential information such as cancer type, initial staging or recurrences, and prior treatment. These details can directly impact protocol choice and diagnostic interpretation. Inadequate indications have persisted despite educational efforts directed at referring providers.

Substantial variability remains in clinicians' understanding of what constitutes radiologically relevant information, and the resultant communication gap is a well-recognized contributor to diagnostic uncertainty and avoidable safety risks. Faced with rising volumes and time pressure, radiologists may be forced to choose between proceeding with only the provided indication or restricting chart reviews to a few notes that can be prone to inaccuracy.

There is a pressing need to develop scalable, automated solutions to improve the quality of provided imaging indications.

Across multiple radiologic applications, previous work has demonstrated the potential of large language models (LLMs) in radiology-relevant tasks, including annotating reports with disease labels (4), producing differential diagnoses (5), and generating radiologic impressions (6). Moreover, LLMs have shown promise in generating and synthesizing clinical text, with prior studies demonstrating the ability of LLMs to summarize progress notes into a condensed list of medical issues and conditions (7), assess the completeness of requisition histories (8), and generate requisition histories for oncologic imaging (9). However, these efforts were largely limited in scope—often restricted to narrow clinical domains, single proprietary models, or simplified extraction tasks—and were not evaluated in a sizable clinical reader study.

This study aimed to evaluate whether LLMs can improve the clinical utility of provided imaging indications by automatically leveraging information from clinical notes. We evaluated on-premises, open-source, and cloud-based proprietary LLMs for their ability to generate radiology-relevant clinical histories from patients' electronic medical records. We then conducted a

Abbreviations

LLM = large language model, SE = standard error

Summary

Proprietary and open-source large language models (LLMs) generated radiologically relevant indications from clinical notes that were more comprehensive and factual than clinician indications, and when generated by the proprietary LLM, were ranked most useful in protocoling and imaging interpretation by radiologists.

Key Results

- In a reader study involving 241 radiology examinations across body systems and pathophysiologic groups, readers rated indications from both the best-performing proprietary (Claude 3.5 Sonnet; Anthropic) and open-source (Qwen 2.5-7B Instruct; Alibaba) large language model (LLM) as more comprehensive and factual than those of the referring clinician (all $P < .001$).
- Compared with the best open-source LLM, referring clinician, and radiologist, the proprietary LLM was ranked the highest in usefulness in protocoling (all $P < .001$), usefulness in interpretation (all $P < .001$), and overall ranking (all $P < .001$).
- Readers cited comprehensiveness (65.77% of ratings; both $P < .001$) as the most important factor influencing overall rankings compared with factuality and conciseness.

clinical reader study involving 20 radiologists to evaluate how LLM-generated clinical indications compared with clinician-provided and radiologist-curated indications across a variety of body systems and pathophysiologic categories for use in image protocoling and interpretation.

Materials and Methods

Study Design

This retrospective study was approved by the institutional review board of the University of California San Francisco (or UCSF), was compliant with the Health Insurance Portability and Accountability Act, and had the requirement for informed consent waived. First, a variety of open-source and proprietary LLMs were evaluated on automated benchmarks. The respective best-performing, open-source, and proprietary model on these benchmarks were selected for the clinical reader performance study.

Data Collection

Data were obtained from the UCSF Information Commons (10), a deidentified clinical data warehouse containing structured and unstructured multimodal electronic health record data from more than 6 million patients (January 2012 to August 2024). Not every imaging examination in this time frame had a radiologist-curated clinical history, as these are optional additions by the interpreting radiologist in our department workflow. Imaging studies were included only if the radiology report contained both the referring provider's clinical history included with the imaging order and the additional radiologist-curated clinical history (Fig 1A).

All clinical notes associated with each patient were extracted across all encounters that occurred before the interpretation date of the imaging examination. Then, certain notes such as telephone encounters, patient instructions, nursing workflow notes, and notes with less than 20 words were excluded (Fig 1B).

Patients with no notes after excluding these notes were excluded. This compiled dataset is referred to as the UCSF Imaging Indication Dataset (Fig 1C).

To evaluate model performance across diverse clinical contexts, we created stratified subsets of the UCSF Imaging Indication Dataset. To account for intrinsic category imbalances in our dataset, we further stratified imaging studies by body system and pathophysiologic categories. For the body systems, we used a keyword-based matching method (Table S1) and for the pathophysiologic categories, we used an LLM-based labeling approach (Appendix S1). We created two stratified subsets balanced across the 25 body system–pathophysiology pairings: one referred to as the LLM selection test set and the other as the reader study test set. Moreover, a randomly sampled subset was used for prompt engineering and hyperparameter tuning referred to as the LLM validation set. Further details regarding dataset stratification can be found in Appendix S1.

For the reader study test set, the examinations were manually reviewed and excluded when key clinical details appeared only in the referring clinician–provided imaging indication, typically because the clinical note was not yet finalized at the time of imaging interpretation, leaving no reference standard for evaluation. Race and ethnicity were self-reported by patients and are presented to describe the study sample and to inform assessment of the generalizability of the findings (Table S1).

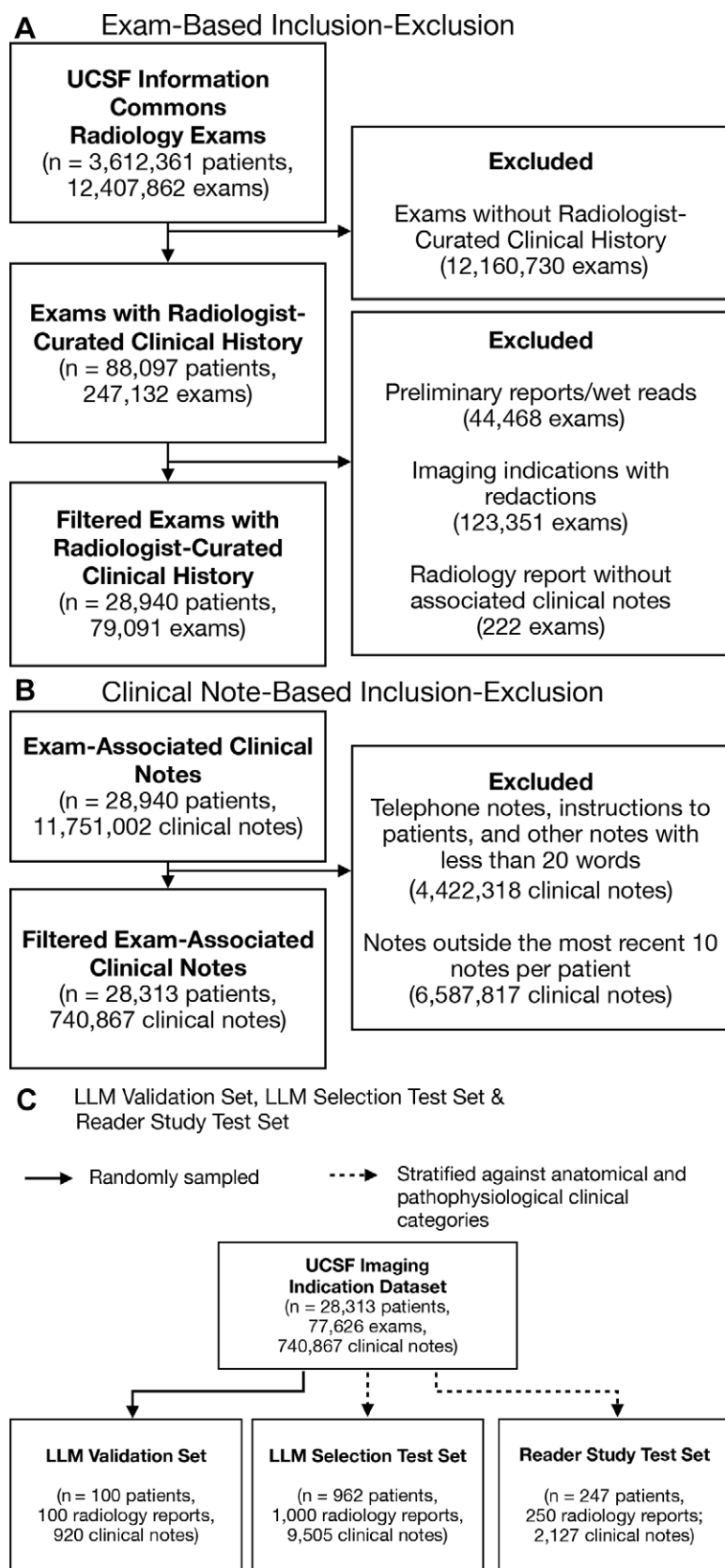
LLM Selection

Multiple proprietary and open-source LLMs, with and without explicit reasoning capabilities, were evaluated (Appendix S2). Each model received the referring clinician–provided indication and the 10 most recent clinical notes and generated an enhanced imaging indication (Fig 2). Notes were included based on the date of the examination, so that notes created after an examination date were not used as input. The number of input notes was fixed to ensure comparability across models, standardize the amount of contextual information provided, and remain within model context limits (Fig S1).

Prompt optimization and hyperparameter tuning were performed on the LLM validation set before benchmarking to improve output quality and reduce variability (Appendix S2, Table S2). Model outputs were evaluated using ROUGE (Recall-Oriented Understudy for Gisting Evaluation) (11), MEDCON (Medical Concept-based Evaluation Metric) (12), RADGRAPH (Radiology Knowledge Graph Entity F1 Score) (13), and BERTScore (Bidirectional Encoder Representations from Transformers Score) (14), with detailed descriptions of each metric provided in Appendix S3. Models were ranked across metrics using mean rank. The proprietary and open-source models with the best mean rank were selected for the reader study.

Clinical Reader Performance Study

A clinical reader performance study was conducted to evaluate the quality of clinical histories generated by the best-performing LLMs. Twenty radiologists were recruited for the reader study, consisting of 14 radiologists in training (average postgraduate year experience, 2.9 years) (Table S3) and six board-certified attending radiologists (average postgraduate year experience, 14 years) (Table S3).



Each radiologist was assigned 25 unique examinations such that each of the 250 examinations was reviewed by two different radiologists. Furthermore, examination assignments maximized diversification of reader pairs by minimizing the number of shared examinations between any given single pair of readers

Figure 1: Inclusion-exclusion flowchart and dataset stratification. **(A)** Radiology report inclusion and exclusion. Reports with radiologist-curated clinical histories were retained after exclusion of preliminary (or wet read) reports, redacted indications, and reports without associated clinical notes. **(B)** Clinical note inclusion and exclusion. Report-associated clinical notes were retained after exclusion of telephone or patient-instruction notes, short notes, and notes outside the 10 most recent per patient. **(C)** Dataset stratification and sampling. After clinical note inclusion-exclusion, radiology reports with no associated clinical notes were excluded. The University of California San Francisco (UCSF) Imaging Indication Dataset was stratified by anatomic and pathophysiologic categories and sampled to construct the large language model (LLM) validation set, LLM selection test set, and reader study test set.

(15). For the purposes of examination assignment, no distinction was made between levels of training.

The study was conducted on a custom-built evaluation platform (Fig S2). For each examination, the radiologists were asked to evaluate four imaging indications: (a) the referring clinician's provided indication; (b) the radiologist-curated indication; (c) the indication generated by the best-performing open-source LLM; and (d) the indication generated by the best-performing proprietary LLM on comprehensiveness, factuality, and conciseness, similar to prior work (7). These were each rated on a Likert-type scale from 1 to 5 with detailed criteria (Fig S3) prepared by two authors (T.L.C., postgraduate year 5 radiology resident, and J.H.S., with 4 years of post-fellowship training), which were provided to each radiologist. The readers were provided with the same 10 most recent clinical notes that were provided as input to the LLMs. The readers were blinded to the source of each indication.

Readers were further asked to rank the indications based on usefulness for selecting an examination protocol, usefulness for imaging interpretation, and overall ranking. We also asked the radiologists to rank which factor (comprehensiveness, factuality, or conciseness) most influenced their overall ranking. Optional text-based feedback was collected on each indication and a general comment on the examination overall.

Statistical Analysis

For the automated LLM selection, pairwise comparisons between each model and the referring clinician were performed using Wilcoxon signed rank tests on paired examination-level scores, with Holm correction within each metric. Subgroup analyses for the best-performing proprietary and open-source LLMs (by body system and pathophysiologic category) used the same approach.

For the clinical reader study, comprehensiveness, factuality, and conciseness were assessed on ordinal Likert scales using cumulative link mixed models with a logit link (proportional odds). Indication source was a fixed effect; examination and reader were included as random intercepts to account for clustering within the incomplete block design. Separate models were fit for each outcome, and effects are reported as log-odds coefficients (logit estimates) relative to the

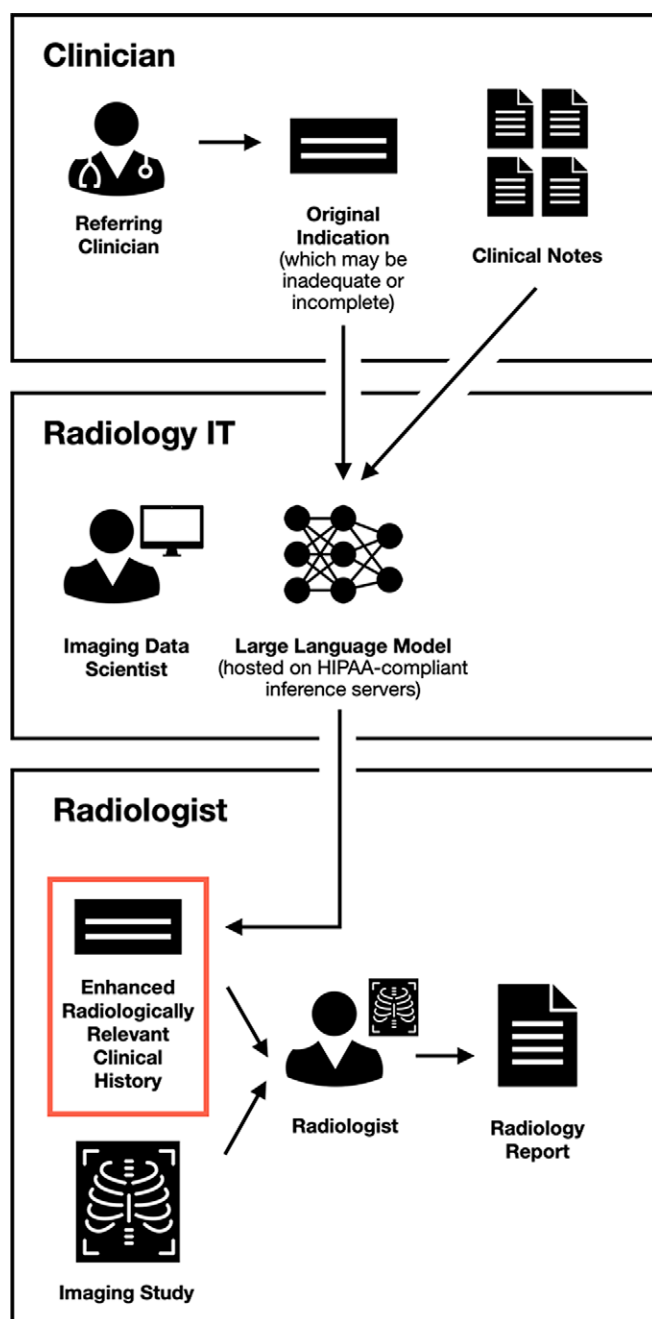


Figure 2: Process flow of the radiologically relevant clinical history summarization pipeline with large language models. Patients' clinical notes and the referring clinician–provided indication serve as inputs into the large language model, within a deployment inference system developed by a radiology information technology (IT) imaging data scientist. The model subsequently generates a more comprehensive indication to support radiologists in imaging interpretation and in their radiology report. HIPAA = Health Insurance Portability and Accountability Act.

referring physician. A separate cumulative link mixed model evaluated factors influencing overall ranking, with rank position as the ordinal outcome, criterion type as the fixed effect, and examination and reader as random intercepts. Post hoc pairwise contrasts used Tukey adjustment (emmeans package, version 2.0.1).

Statistical significance was set at a *P* value of less than .05. Analyses were conducted in R (version 4.5.2; R Foundation for Statistical Computing) with the ordinal package (version 2025.12.29) and emmeans package (A.S., T.L.C.) (16). All

code is made available and may be accessed at <https://github.com/bdrad/llm-radiological-clinical-history-summarization>.

Results

Patient Characteristics

This study included 28313 patients (mean age, 59 years $\pm$ 20.6 [SD]; age range, 2–91 years; 14912 women) (Table 1). Patients were excluded based on examination-level and clinical note-level criteria (Fig 1). At the examination level, exclusions included radiology examinations without radiologist-curated clinical history ($n = 12\,160\,730$), preliminary (or wet read) examinations ($n = 44\,468$), examinations with redacted imaging indications ($n = 123\,351$), and reports without associated clinical notes ($n = 222$). At the clinical note level, exclusions included telephone notes, instructions to patients, and other notes with less than 20 words ($n = 4\,422\,318$), as well as notes outside the 10 most recent notes per patient ($n = 6\,587\,817$) (Figs 1A, 1B). After exclusion, the UCSF Imaging Indication Dataset was composed of 77626 unique examinations and 740867 unique clinical notes across all patients. For the LLM selection test set, a subset of 1000 stratified radiology report samples were collected from the UCSF Imaging Indication Dataset for each combination of the five body systems and five pathophysiologic categories, resulting in 40 examinations for each of the 25 body system–pathophysiologic pairs.

Following the same stratified sampling approach, we selected 250 examinations for the reader study test set, with 10 examinations sampled from each body system–pathophysiologic category pair. Before analysis, nine examinations were excluded because key clinical details appeared only in the referring clinician–provided imaging indication and were absent from the clinical notes. Because these details could not be verified using the available reference information, these examinations were excluded from the final analysis. The analytic reader study test set included 241 examinations, each evaluated independently by two radiologists, yielding 482 reader-examination evaluations.

LLM Pathophysiologic Category Labeling Evaluation

In the manual evaluation of the results of the automated pathophysiologic labeling by an author (T.L.C.), GPT-4o (OpenAI) achieved an overall accuracy of 93.00% (93 of 100 examinations) on the LLM Validation Set (Appendix S1). This was considered adequate for purposes of stratified sampling for the study.

LLM Selection and Automated Evaluation

Figure 3 demonstrates the performance of the various LLMs across the four different quantitative evaluation metrics. Sample imaging indications generated by the different models are shown in Table S4. Claude 3.5 Sonnet (Anthropic) was the best-performing model across both proprietary and open-source models (Tables S5, S6): Claude achieved a mean ROUGE score of 20.61 (95% CI: 19.69, 21.58), MEDCON score of 21.21 (95% CI: 19.77, 22.70), RADGRAPH score of 9.72 (95% CI: 8.55, 10.91), and BERTScore of 32.37 (95% CI: 31.27, 33.44). Among the open-source models, Qwen 2.5-7B Instruct (Alibaba) was the best-performing model, competitive with the other proprietary models (Tables S5, S6):

Table 1: Patient Demographics and Examination-Level Characteristics of the Clinical Indication Datasets

Characteristic	All Patients with Radiology Reports (<i>n</i> = 3 612 361)	UCSF Imaging Indication Dataset (<i>n</i> = 28 313)	LLM Validation Set (<i>n</i> = 100)	LLM Selection Test Set (<i>n</i> = 962)	Reader Study Test Set (<i>n</i> = 247)
Total no. radiology reports	12 407 862	77 626	100	1000	250
Total no. clinical notes excluding radiology reports	216 811 438	740 867	920	9505	2127
Age (y)*	49 ± 25.8 (1–91)	59 ± 20.6 (2–91)	57 ± 20.1 (4–91)	57 ± 20.6 (3–91)	57 ± 19.6 (3–91)
Sex					
Male	1 638 764 (45.37)	13 269 (46.87)	49 (49.00)	451 (46.88)	122 (49.39)
Female	1 968 086 (54.48)	14 912 (52.67)	49 (49.00)	509 (52.91)	124 (50.20)
Other†	5511 (0.15)	132 (0.47)	2 (2.00)	2 (0.21)	1 (0.40)
Race and ethnicity					
Asian	318 755 (8.82)	5321 (18.79)	16 (16.00)	184 (19.13)	46 (18.62)
Black/African American	198 699 (5.50)	2094 (7.40)	6 (6.00)	87 (9.04)	19 (7.69)
Native American/Alaska Native	14 091 (0.39)	227 (0.80)	2 (2.00)	9 (0.94)	0 (0.00)
Native Hawaiian/Pacific Islander	107 532 (2.98)	245 (0.87)	0 (0.00)	5 (0.52)	0 (0.00)
White	1 485 460 (41.12)	15 167 (53.57)	56 (56.00)	510 (53.01)	126 (51.01)
Other†	1 487 824 (41.19)	5259 (18.57)	20 (20.00)	167 (17.36)	56 (22.67)
Imaging modality					
MRI	1 454 877 (11.73)	38 163 (49.16)	48 (48.00)	307 (30.70)	65 (26.00)
CT	2 071 226 (16.69)	35 807 (46.13)	46 (46.00)	584 (58.4)	153 (61.20)
US	1 434 143 (11.56)	1866 (2.40)	3 (3.00)	73 (7.30)	21 (8.40)
Radiography	4 194 542 (33.81)	1003 (1.29)	1 (1.00)	31 (3.10)	7 (2.80)
Other†	3 253 074 (26.22)	787 (1.02)	2 (2.00)	5 (0.50)	4 (1.60)
Body system					
Brain	1 015 726 (8.19)	53 970 (69.53)	67 (67.00)	200 (20.00)	50 (20.00)
Head/neck	321 852 (2.59)	9750 (12.56)	16 (16.00)	200 (20.00)	50 (20.00)
MSK	1 658 211 (13.36)	5592 (7.20)	10 (10.00)	200 (20.00)	50 (20.00)
Abdomen/pelvis	1 952 072 (15.73)	3956 (5.10)	5 (5.00)	200 (20.00)	50 (20.00)
Chest	2 389 029 (19.25)	2899 (3.73)	0 (0.00)	200 (20.00)	50 (20.00)
Other†	5 070 972 (40.87)	1459 (1.88)	2 (2.00)	0 (0.00)	0 (0.00)
Pathophysiologic category					
Cancer/mass	...	24 573 (31.66)	35 (35.00)	200 (20.00)	50 (20.00)
Symptom-based	...	24 258 (31.25)	34 (34.00)	200 (20.00)	50 (20.00)
Surgical	...	15 050 (19.39)	20 (20.00)	200 (20.00)	50 (20.00)
Infection/inflammatory	...	8826 (11.37)	5 (5.00)	200 (20.00)	50 (20.00)
Structural	...	4906 (6.32)	6 (6.00)	200 (20.00)	50 (20.00)
Other†	...	13 (0.01)	0 (0.00)	0 (0.00)	0 (0.00)

Note.—Data are presented as numbers with percentages in parentheses, unless otherwise indicated. Patients comprising the clinical indication datasets were sourced from the University of California San Francisco (UCSF) Information Commons. The large language model (LLM) validation set was used to evaluate pathophysiologic labels, perform prompt engineering, and run temperature ablation experiments. Data were stratified by anatomy and pathophysiologic categories to create the LLM selection test set and reader study test set for evaluating LLM performance across diverse clinical contexts. Pathophysiologic categorization was performed only on the UCSF Imaging Indication Dataset and its derived subsets (LLM validation set, LLM selection test set, and reader study test set) due to feasibility and cost considerations. MSK = musculoskeletal.

* Data are means ± SDs, with ranges in parentheses.

† The “Other” sex category includes nonbinary patients or those who did not report sex. The “Other” race and ethnicity category includes patients who do not identify with the specified race and ethnicity groups or who did not report their race and ethnicity. The “Other” imaging modality category includes mammography, dual-energy x-ray absorptiometry bone density, fluoroscopy, and additional modalities not categorized as MRI, CT, US, or radiography in this study. The “Other” body system category denotes imaging examinations not classified into the predefined keyword-based body region categories. The “Other” pathophysiologic category denotes imaging examinations not classified into the predefined pathophysiologic categories.

Qwen achieved a mean ROUGE score of 18.31 (95% CI: 17.46, 19.18), MEDCON score of 19.81 (95% CI: 18.43, 21.22), RADGRAPH score of 7.51 (95% CI: 6.48, 8.58), and BERTScore of 30.25 (95% CI: 29.23, 31.26). Aside from

DeepSeek R1 Distill Qwen 2.5-7B and Biomistral (Avignon University and Nantes University, respectively), all the LLMs outperformed the referring clinician across all metrics (all *P* < .001) (Table S7).

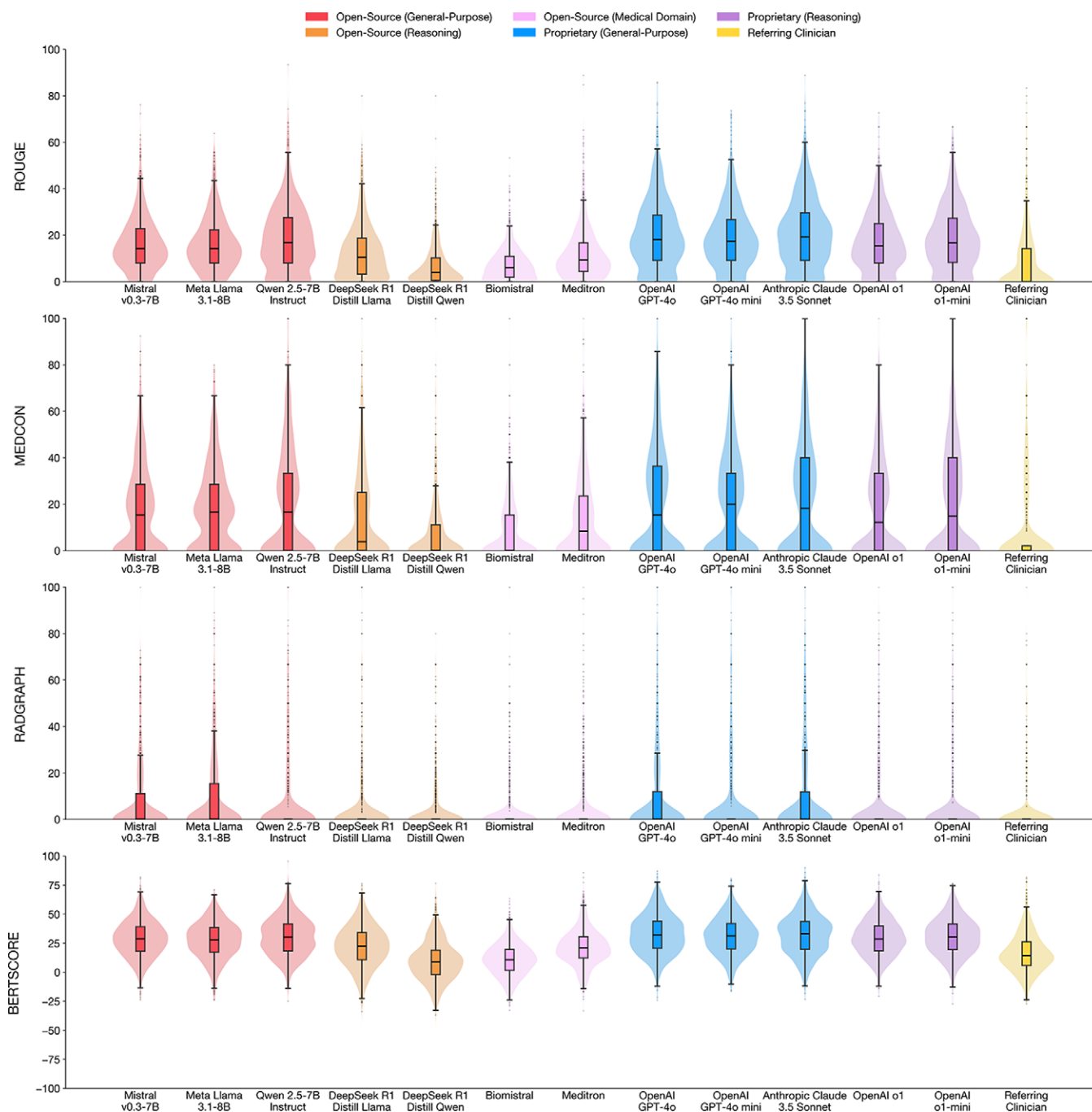


Figure 3: Automated evaluation of large language model performance against the radiologist-curated indication. Open-source models (Mistral v0.3, Mistral AI; Llama 3.1-8B-Instruct, Meta; Qwen 2.5-7B Instruct, Alibaba; DeepSeek Llama-3.1-8B-Instruct, DeepSeek; DeepSeek Qwen 2.5-7B Instruct, DeepSeek; Llama-3.1 Meditron-3-8B and Biomistral, Avignon University and Nantes University, respectively), proprietary models (GPT-4o, OpenAI; GPT-4o mini, OpenAI; Claude 3.5 Sonnet, Anthropic; GPT-o1, OpenAI; and GPT-o1 mini, OpenAI), and the referring clinician were measured against the radiologist-curated indication using the ROUGE, MEDCON, RADGRAPH, and BERTSCORE automated evaluation metrics. Distributions are shown as violin plots, in which the horizontal spread reflects the kernel density of the scores. Overlaid box plots indicate the medians (center line) and IQRs (box), with whiskers showing the nonoutlier data ranges and points denoting outliers. Colors denote model categories: red, open-source general-purpose; orange, open-source reasoning; pink, open-source medical-domain; blue, proprietary general-purpose; purple, proprietary reasoning; yellow, referring clinician. Across most automated evaluation metrics, large language model-generated indications outperformed referring clinician indications. BERTSCORE = Bidirectional Encoder Representations from Transformers Score, MEDCON = Medical Concept-based Evaluation Metric, RADGRAPH = Radiology Knowledge Graph Entity F1 Score, ROUGE = Recall-Oriented Understudy for Gisting Evaluation.

The best-performing proprietary (Claude 3.5 Sonnet) and the best-performing open-source model (Qwen 2.5-7B Instruct) were also evaluated against the referring clinicians' performance across body systems and pathophysiologic categories (Fig 4). Across most evaluation metrics, body systems, and pathophysiologic categories, both LLMs outperformed the referring clinician; exceptions were observed for Qwen 2.5-7B

Instruct in symptom-based and structural pathophysiologic groups (Table S8).

Tables 2 and S9 include sample outputs from the best-performing proprietary LLM, best-performing open-source model, alongside the referring clinician-provided indication and the radiologist-curated indication across pathophysiologic categories and body systems, respectively.

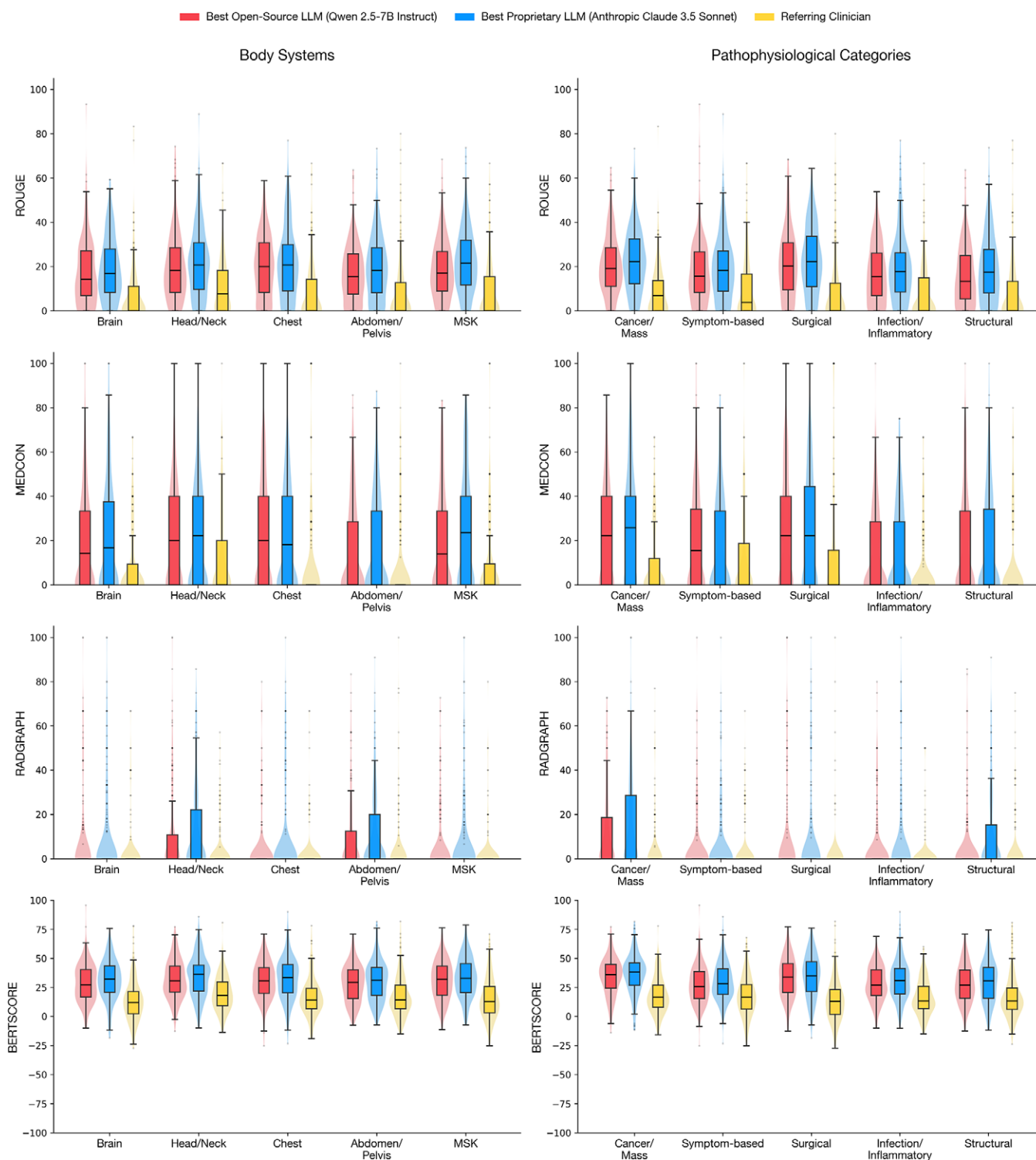


Figure 4: Automated evaluation of the best-performing large language models (LLMs) and the referring clinician's indications against the radiologist-curated indication. The ROUGE, MEDCON, RADGRAPH, and BERTSCORE automated evaluation metrics were calculated across different clinical contexts for the best-performing proprietary LLM (Claude 3.5 Sonnet; Anthropic), and the best-performing open-source LLM (Qwen 2.5-7B Instruct; Alibaba), and the referring clinician. Across most evaluation metrics, body systems, and pathophysiologic categories, both LLMs outperformed the referring clinician; exceptions were observed for Qwen 2.5-7B Instruct in symptom-based and structural pathophysiologic groups. BERTSCORE = Bidirectional Encoder Representations from Transformers Score, MEDCON = Medical Concept-based Evaluation Metric, MSK = musculoskeletal, RADGRAPH = Radiology Knowledge Graph Entity F1 Score, ROUGE = Recall-Oriented Understudy for Gisting Evaluation.

Clinical Reader Study

Cumulative link mixed models were fitted for all ordinal outcomes to evaluate the effect of indication source (referring clinician–provided, radiologist-curated, Claude 3.5 Sonnet, and Qwen 2.5-7B Instruct). Outcomes included Likert ratings of comprehensiveness,

factuality, and conciseness, and ordinal rankings of usefulness for protocoling, interpretation, and overall ranking. Models included random intercepts for readers ($n = 20$) and examinations ($n = 241$). Post hoc pairwise contrast tests assessed differences between sources (Tables 3, 4). A separate cumulative link mixed model evaluated

Table 2: Sample Indications Across Different Pathophysiologic Categories

Category and Clinical Information	Examination Type	Indication from Referring Clinician	Indication from Radiologist	Indication from Best Open-Source LLM (Qwen 2.5-7B Instruct; Alibaba)	Indication from Best Proprietary LLM (Claude 3.5 Sonnet; Anthropic)
Cancer/mass (Male, 60 y)	CT NECK WITH CONTRAST	Other reasons than neck pain or neck trauma preop cancer staging cSCC face, surgery planning and/or restaging; has concurrent rectal cancer.	Slowly growing left cheek mass, path proven SCC	Left facial cSCC, postparotidectomy and neck dissection, with rectal cancer, preop staging.	Male with facial cSCC, s/p left parotidectomy and neck dissection September 13, 2020, for preop staging.
Symptom-based (Female, 72 y)	MR IAC TEMPORAL BONE WITHOUT AND WITH CONTRAST	Asymmetric hearing loss	Right-sided tinnitus. Asymmetric hearing loss, right greater than left.	Asymmetric tinnitus and hearing loss in a 72-year-old female.	72-year-old female with right-sided tinnitus and asymmetric hearing loss.
Surgical (Female, 57 y)	CT CHEST WITHOUT CONTRAST	S/p lung transplant	Status post uncomplicated bilateral lung transplant on April 24, 2022. Donor RML wedge resection showing typical carcinoid tumor without clear margins s/p returned OR May 7 for right VATS for clear margins.	57-year-old female postbilateral lung transplant, recovering from RML wedge resection for donor carcinoid tumor.	57-year-old female s/p bilateral lung transplant on April 24, 2022, recent VATS for donor carcinoid tumor.
Infection/inflammatory (Male, 57 y)	MR CERVICAL SPINE WITH AND WITHOUT CONTRAST, MR THORACIC SPINE WITH AND WITHOUT CONTRAST	MS	Continue off DMT for MS with clinical and MRI monitoring.	58-year-old male with relapsing remitting MS, lymphopenia, last seen June 2022, stable on hydroxychloroquine and minoxidil.	57-year-old male with relapsing-remitting MS, off DMT since 2020, with recent bladder incontinence.
Structural (Male, 14 y)	MR BRAIN/MR ANGIOGRAM WITH AND WITHOUT CONTRAST	Cerebral artery aneurysm or AVM suspected	Unruptured large thalamic AVM (SM 4-eloquent, 5 cm, deep venous drainage). Started treatment with trametinib (MEK inhibitor) in October 2022.	Unruptured brain AVM, started trametinib October 2022, scheduled MRI next week.	15-year-old male with unruptured right thalamic AVM, on trametinib since October 2022.

Note.—Indications provided by the referring clinician, radiologist, and the corresponding outputs from the best open-source and best-proprietary model are provided for different pathophysiologic categories. Minor age discrepancies reflect deidentification-related date shifting or differences between age documented at the time of clinical note entry and age at the time of imaging. The dates as displayed in this table are automatically shifted by random increments for anonymity through the deidentification process. The indications generated by the large language models (LLMs) were qualitatively more comprehensive and clinically relevant than those provided by referring clinicians and were comparable in quality to radiologist-curated indications. AVM = arteriovenous malformation, cSCC = cutaneous squamous cell carcinoma, DMT = disease modifying therapy, IAC = internal auditory canal, MEK = mitogen-activated protein kinase, MS = multiple sclerosis, OR = operating room, preop = preoperative, RML = right middle lobe, SM 4 = Spetzler-Martin grade 4, s/p = status post, SCC = squamous cell carcinoma, VATS = video-assisted thoracoscopic surgery.

Table 3: Tukey-adjusted Pairwise Contrasts Between Indication Sources from Cumulative Link Mixed Models for Comprehensiveness, Factuality, and Conciseness

Source A vs Source B	Logit Estimate	Standard Error	P Value
Comprehensiveness			
Referring clinician vs radiologist	-1.42	0.12	<.001
Referring clinician vs Claude 3.5 Sonnet	-2.55	0.13	<.001
Referring clinician vs Qwen 2.5-7B Instruct	-2.19	0.13	<.001
Radiologist vs Claude 3.5 Sonnet	-1.14	0.12	<.001
Radiologist vs Qwen 2.5-7B Instruct	-0.78	0.12	<.001
Claude 3.5 Sonnet vs Qwen 2.5-7B Instruct	0.36	0.12	.011
Factuality			
Referring clinician vs radiologist	-0.38	0.13	.021
Referring clinician vs Claude 3.5 Sonnet	-1.10	0.14	<.001
Referring clinician vs Qwen 2.5-7B Instruct	-0.66	0.13	<.001
Radiologist vs Claude 3.5 Sonnet	-0.72	0.14	<.001
Radiologist vs Qwen 2.5-7B Instruct	-0.27	0.14	.19
Claude 3.5 Sonnet vs Qwen 2.5-7B Instruct	0.45	0.14	.009
Conciseness			
Referring clinician vs radiologist	0.70	0.14	<.001
Referring clinician vs Claude 3.5 Sonnet	0.093	0.14	.91
Referring clinician vs Qwen 2.5-7B Instruct	0.93	0.14	<.001
Radiologist vs Claude 3.5 Sonnet	-0.61	0.13	<.001
Radiologist vs Qwen 2.5-7B Instruct	0.22	0.13	.29
Claude 3.5 Sonnet vs Qwen 2.5-7B Instruct	0.83	0.13	<.001

Note.—Higher Likert ratings indicate better performance (1 = lowest; 5 = highest). Logit estimates represent differences in the odds of source A receiving a higher rating relative to source B. Negative estimates indicate lower odds for source A of receiving higher ratings relative to source B, whereas positive estimates indicate higher odds. Both the best-performing proprietary (Claude 3.5 Sonnet; Anthropic) and open-source (Qwen 2.5-7B Instruct; Alibaba) large language models were rated as more comprehensive and factual than the referring clinician.

Table 4: Tukey-adjusted Pairwise Contrasts Between Indication Sources from Cumulative Link Mixed Models for Usefulness in Protocols, Usefulness in Interpretation, and Overall Ranking

Source A vs Source B	Logit Estimate	Standard Error	P Value
Usefulness for protocols			
Referring clinician vs radiologist	0.36	0.12	.016
Referring clinician vs Claude 3.5 Sonnet	1.93	0.12	<.001
Referring clinician vs Qwen 2.5-7B Instruct	1.36	0.12	<.001
Radiologist vs Claude 3.5 Sonnet	1.57	0.12	<.001
Radiologist vs Qwen 2.5-7B Instruct	1.00	0.12	<.001
Claude 3.5 Sonnet vs Qwen 2.5-7B Instruct	-0.57	0.11	<.001
Usefulness for interpretation			
Referring clinician vs radiologist	0.80	0.12	<.001
Referring clinician vs Claude 3.5 Sonnet	2.36	0.13	<.001
Referring clinician vs Qwen 2.5-7B Instruct	1.66	0.12	<.001
Radiologist vs Claude 3.5 Sonnet	1.56	0.12	<.001
Radiologist vs Qwen 2.5-7B Instruct	0.86	0.12	<.001
Claude 3.5 Sonnet vs Qwen 2.5-7B Instruct	-0.70	0.12	<.001
Overall ranking			
Referring clinician vs radiologist	0.69	0.12	<.001
Referring clinician vs Claude 3.5 Sonnet	2.29	0.13	<.001
Referring clinician vs Qwen 2.5-7B Instruct	1.64	0.12	<.001
Radiologist vs Claude 3.5 Sonnet	1.60	0.12	<.001
Radiologist vs Qwen 2.5-7B Instruct	0.95	0.12	<.001
Claude 3.5 Sonnet vs Qwen 2.5-7B Instruct	-0.65	0.12	<.001

Note.—Lower rank values indicate greater preference (rank 1 = most useful). Logit estimates represent differences in the odds of source A receiving a worse ranking relative to source B. Positive estimates indicate higher odds that source A received worse rankings than source B. Compared with the best-performing open-source large language model, referring clinician, and radiologist, the proprietary large language model was ranked as more useful for protocols, interpretation, and overall ranking. Claude 3.5 Sonnet, Anthropic; Qwen 2.5-7B Instruct, Alibaba.

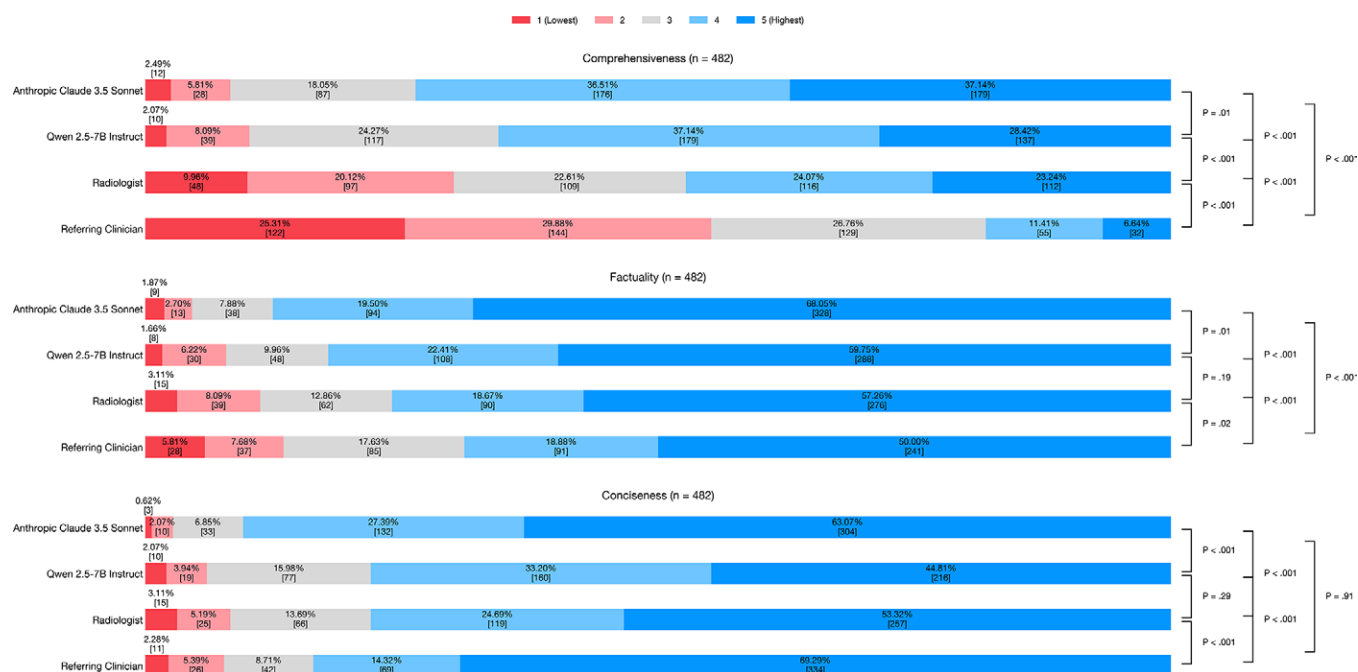


Figure 5: Likert ratings of comprehensiveness, factuality, and conciseness in the clinical reader study. Stacked bar charts show the percentage distribution of ratings (1 = lowest; 5 = highest) assigned to imaging indications from the referring clinician, radiologist, best-performing proprietary large language model (LLM) (Claude 3.5 Sonnet; Anthropic), and the best-performing open-source LLM (Qwen 2.5-7B Instruct; Alibaba) across comprehensiveness, factuality, and conciseness ($n = 482$ ratings). LLM-generated indications received higher ratings for comprehensiveness and factuality than the referring clinician-provided indications, with the proprietary model showing the greatest proportion of top ratings.

factors influencing the overall ranking, with rank position as the ordinal outcome, factor as the fixed effect, and random intercepts for readers and examinations. Post hoc pairwise contrast tests assessed differences between factors (Table S10).

Likert ratings are shown in Figure 5. For comprehensiveness, Claude 3.5 Sonnet received the highest percentage of top ratings (Likert rating of 5: 37.14% [179 of 482]), which was followed by Qwen 2.5-7B Instruct (28.42% [137 of 482]), the radiologist (23.24% [112 of 482]), and the referring clinician (6.64% [32 of 482]). The referring clinician received lower ratings than both Claude (logit estimate: -2.55 ; standard error [SE]: 0.13; $P < .001$) and Qwen (logit estimate: -2.19 ; SE: 0.13; $P < .001$). The radiologist also received lower ratings than both models (Claude, logit estimate: -1.14 ; SE: 0.12; Qwen, logit estimate: -0.78 ; SE: 0.12; both $P < .001$).

For factuality, Claude similarly received the highest percentage of top ratings (Likert rating of 5: 68.05% [328 of 482]), which was followed by Qwen (59.75% [288 of 482]), the radiologist (57.26% [276 of 482]), and the referring clinician (50.00% [241 of 482]). The referring clinician received lower ratings than both Claude (logit estimate: -1.10 ; SE: 0.14; $P < .001$) and Qwen (logit estimate: -0.66 ; SE: 0.13; $P < .001$). The radiologist received lower ratings than Claude (logit estimate: -0.72 ; SE: 0.14; $P < .001$), but there was no evidence of a difference in factuality scores between the radiologist and Qwen (logit estimate: -0.27 ; SE: 0.14; $P = .19$).

For conciseness, the referring clinician received the highest percentage of top ratings (Likert rating of 5: 69.29% [334 of 482]), which was followed by Claude (63.07% [304 of 482]), the radiologist (53.32% [257 of 482]), and Qwen (44.81% [216 of 482]). There was no evidence of a difference in conciseness scores between the referring clinician and Claude (logit estimate: 0.094; SE: 0.14; $P = .91$), but the referring clinician received higher

ratings than Qwen (logit estimate: 0.93; SE: 0.14; $P < .001$). The radiologist's indications received lower ratings than Claude (logit estimate: -0.61 ; SE: 0.13; $P < .001$), whereas the conciseness scores between Qwen and the radiologist were similar (logit estimate: 0.22; SE: 0.13; $P = .29$).

The readers' rankings of the indications based on their usefulness for protocoling, usefulness for interpretation, and overall ranking are illustrated in Figure 6A. In terms of usefulness for protocoling, Claude received the highest percentage of top rankings (rank 1: 40.87% [197 of 482]), which was followed by Qwen (26.35% [127 of 482]), the radiologist (21.58% [104 of 482]), and the referring clinician (11.20% [54 of 482]). Claude outranked Qwen Instruct, the radiologist, and the referring clinician (all $P < .001$).

In terms of usefulness for interpretation, Claude also received the highest percentage of top rankings (rank 1: 44.61% [215 of 482]), which was followed by Qwen (24.48% [118 of 482]), the radiologist (24.27% [117 of 482]), and the referring clinician (6.64% [32 of 482]). Claude 3.5 Sonnet similarly outranked Qwen 2.5-7B Instruct, the radiologist, and the referring clinician (all $P < .001$).

In terms of overall ranking, Claude again received the highest percentage of top rankings (rank 1: 44.19% [213 of 482]), which was followed by Qwen (25.93% [125 of 482]), the radiologist (23.03% [111 of 482]), and the referring clinician (6.85% [33 of 482]). Claude 3.5 Sonnet outranked Qwen 2.5-7B Instruct, the radiologist, and the referring clinician (all $P < .001$).

Error and Safety Analysis

When asked which factor most influenced their overall rankings, readers cited comprehensiveness (65.77% [317 of 482]; both $P < .001$) most often as opposed to factuality (29.67% [143 of 482]) and conciseness (4.56% [22 of 482]) (Fig 6B, Table S10).

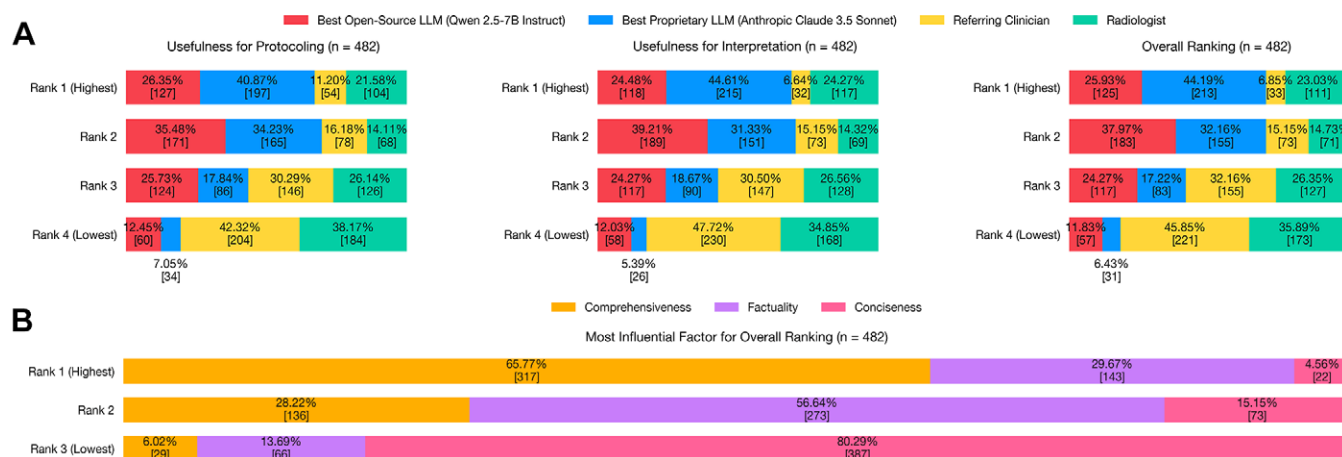


Figure 6: Radiologists' rankings of usefulness for protocoling, usefulness for interpretation, overall ranking, and factors influencing the overall ranking. **(A)** Radiologists' rankings of usefulness for protocoling, usefulness for interpretation, and overall ranking. Stacked bar charts show the distribution of rankings among the referring clinician, radiologist, best proprietary model, and best open-source model based on usefulness for protocoling, usefulness for interpretation, and overall ranking (rank 1 = highest; rank 4 = lowest). The best open-source large language model (LLM) (Qwen 2.5-7B Instruct; Alibaba) is shown in red, the best proprietary LLM (Claude 3.5 Sonnet; Anthropic) in blue, the referring clinician in yellow, and the radiologist in green. Across tasks, the proprietary LLM most frequently received top rankings, followed by the open-source LLM, whereas the referring clinician–provided indications were more often ranked lower overall. **(B)** Radiologists' rankings of factors influencing the overall ranking. The most influential factor contributing to the overall ranking by readers is shown in a stacked bar plot, among comprehensiveness (yellow), factuality (purple), and conciseness (pink) (rank 1 = highest; rank 3 = lowest). Comprehensiveness was most frequently ranked as the top factor contributing to overall ranking, whereas conciseness was less often selected as the top factor.

This emphasis on comprehensiveness was consistent with a review of reader comments on examinations (Table S11), which most frequently highlighted missing clinical details (eg, omission of symptoms, prior radiation therapy, or surgical history) as reasons for lower ratings. Readers also noted occasional factually incorrect information, which included incorrect patient age and incorrect spinal levels. Notably, several comments mentioned that certain clinical information in the indication (eg, exact patient age) could not be reliably verified due to date redactions or was not contained in the provided notes. Examinations in which indications, including the referring clinician–provided and radiologist–curated indication, could not be verified by the provided notes were removed from analysis, as there was no reference standard to rate against.

Discussion

Clinicians face increasingly busy workflows and often do not have sufficient time to write thorough clinical indications when ordering imaging. However, incomplete clinical information can lead to suboptimal imaging interpretations and protocols, potentially leading to patient safety issues (1–3). As such, there is a critical unmet clinical need to develop scalable, automated solutions to improve the quality of provided imaging indications. Our study evaluated whether large language models (LLMs) can improve the clinical utility of provided imaging indications, by leveraging clinical notes, across different body parts and pathophysiologic categories. In our reader study, readers rated indications from both the best-performing proprietary (Claude 3.5 Sonnet; Anthropic) and open-source (Qwen 2.5-7B Instruct; Alibaba) LLM as more comprehensive and factual than those of the referring clinician (all $P < .001$). However, in terms of conciseness, the referring clinician was rated as more concise than Qwen 2.5-7B Instruct ($P < .001$), and there was no evidence of a difference in conciseness scores between Claude 3.5 Sonnet and the referring clinician ($P = .91$). Compared with the best open-source LLM, referring clinician, and radiologist, the proprietary LLM ranked most useful in protocoling (rank 1: 40.87%; all $P < .001$), useful

in interpretation (44.61%; all $P < .001$), and overall ranking (44.19%; all $P < .001$).

An important implication of our findings is that radiologists found LLM-generated indications useful for both interpretation and protocol selection, or at least more so than the referring clinician's indication. This effect was observed across body systems and pathophysiologic categories with both open-source and proprietary models outperforming the indication provided by the ordering clinician. This showcases the potential of LLMs beyond a simple summary of clinical notes, but also as a potential solution to address multiple persistent pain points in the radiologic workflow from protocol prescription to diagnostic interpretation.

It is important to contextualize our study findings within a rapidly evolving technologic landscape. The specific proprietary and open-source models evaluated in this study represented the state of the field at the time of data collection; however, newer, likely more capable LLMs have since been released and will likely continue to be released. As newer and more capable LLMs continue to be released, careful model selection, validation, and compliance review will be essential. We present a scalable, reproducible framework for evaluating LLM-generated imaging indications that can be reapplied to new models. We expect that future models will achieve performance equal to or greater than current ones.

It is essential to address the risk of hallucinations when deploying generative artificial intelligence systems within radiology information technology infrastructure (17). These models may generate inaccurate information, which may adversely affect clinical care. Therefore, it is necessary to clinically validate these artificial intelligence systems on local data before implementation, build in transparency, and have safety guardrails before they are fully integrated into radiologic practice (18). Continuous postimplementation monitoring of outputs is also necessary. Retrospective and prospective clinical reader studies, including thorough education of radiologists using the tool, provide a pathway to mitigate these issues by helping understand where models tend to fail.

LLMs have been previously used to summarize clinical notes, but few studies have conducted comprehensive reader studies to evaluate their ability to summarize radiologically relevant information from unstructured electronic health record data. Van Veen et al (7) focused on generating problem lists from progress notes, Larson et al (8) assessed clinical history completeness with limited detail extraction, and Bhayana et al (9) used GPT-4 for oncologic imaging histories but only within a single model and pathologic condition. In contrast, our study conducts a large-scale reader evaluation benchmarking both open-source and proprietary models against clinicians and radiologists across diverse body systems and pathologic conditions.

Our study has several limitations. First, redactions in the dataset occasionally removed pertinent information (eg, patient age), potentially leading to misclassification of factuality errors, ultimately underestimating model and/or reader performance. Second, for quantitative metrics, we used the “Additional History” section as the reference standard; however, its use is varied in our institution, with some radiologists rewriting full histories, some only supplementing key missing details from the referring clinician-provided history, and others not filling out the field at all. The lattermost was excluded from our study. Third, radiographs and US examinations were underrepresented in our dataset, as radiologists at our institution less often filled out an “Additional History” field when reporting these studies. However, based on included examinations, we expect performance would be like that seen with CT and MRI, because the core question of “Why is this study indicated?” is modality independent. Last, restricting context to the most recent 10 notes limited available information, which made evaluation difficult or impossible in a minority of examinations. Future studies could mitigate this by incorporating broader context windows.

In conclusion, large language models (LLMs) generated more comprehensive, clinically useful imaging indications and were preferred over those provided by referring clinicians. Ultimately, our findings suggest that LLMs may not only solve the longstanding problem of incomplete imaging indications but also help usher in a new era in radiology in which LLMs actively serve as intelligent partners in care, assisting radiologists with imaging protocols, interpretation, and safety measures in real time.

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